6. ESTIMATION OF ASCORBIC ACID (VITAMIN C)

Aim:

To estimate the vitamin C (ascorbic acid) content in the given test solution colorimetrically.

Reagents:

1. **Indophenol reagent:** 20mg of 2,6–dichlorophenolindophenols was dissolved in 10ml of warm distilled water.
2. **DNPH-Thiourea mix reagent:** 2gm of 2,4 – dinitrophenyl hydrazine and 1gm of Thiourea was dissolved in 100ml of 9N sulphuric acid.
3. **5% TCA:** 5gm of Tricholoroacetic acid was dissolved in 100ml of distilled water.
4. **85% sulphuric acid:** 85ml of concentrated sulphuric acid is made up to 100 ml with distilled water.
5. **Stock solution:** 100mg of ascorbic acid dissolved in 100ml of distilled water.
6. **Working standard solution:** 10ml of the stock is diluted to 100ml using distilled water.
   Concentration = 100 µg / ml.

Procedure:

0.2-1.0ml of working standard ascorbic acid solution is pipetted out into five different test tubes and labelled as S1-S5 with the concentration of 20-100µg. 1.0ml of the test solution is pipetted out into a test tube and labelled as T_1. Final volume is made upto 1.0ml using distilled water in all the tubes. A blank is also prepared simultaneously comprising of 1.0ml of distilled water. .0.1ml of indophenol reagent is added to all the test tubes followed by the addition of 1.0ml of DNPH reagent. The contents in the test tubes are mixed well and kept in the boiling water bath for 30 minutes. The test tubes are then cooled and 3.0ml of H_2SO_4 is added with constant shaking. The intensity of red colour developed is measured at 540nm using a colorimeter.

A standard graph is drawn with optical density in Y- axis vs. concentration of ascorbic acid in X axis. The amount of ascorbic acid present in the given test solution is then calculated.

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